

**SET-5****Series %BAB%**प्रश्न-पत्र कोड
Q.P. Code**65/B/5**

रोल नं.

Roll No.

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परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें।

Candidates must write the Q.P. Code on the title page of the answer-book.

- कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ 7 हैं।
- प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें।
- कृपया जाँच कर लें कि इस प्रश्न-पत्र में 14 प्रश्न हैं।
- कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में प्रश्न का क्रमांक अवश्य लिखें।
- इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है। प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा। 10.15 बजे से 10.30 बजे तक छात्र केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे।
- Please check that this question paper contains 7 printed pages.
- Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- Please check that this question paper contains 14 questions.
- **Please write down the serial number of the question in the answer-book before attempting it.**
- 15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer-book during this period.

**गणित**

(केवल दृष्टिबाधित परीक्षार्थियों के लिए)

MATHEMATICS**(FOR VISUALLY IMPAIRED CANDIDATES ONLY)**

निर्धारित समय : 2 घण्टे

Time allowed : 2 hours

अधिकतम अंक : 40

Maximum Marks : 40

65/B/5

Page 1

P.T.O.



General Instructions :

Read the following instructions very carefully and strictly follow them :

- (i) This question paper contains **three** sections – **Section A, B and C**.
- (ii) Each section is **compulsory**.
- (iii) **Section A** has **6** short answer type I questions of **2** marks each.
- (iv) **Section B** has **4** short answer type II questions of **3** marks each.
- (v) **Section C** has **4** long answer type questions of **4** marks each.
- (vi) There is an internal choice in some questions.
- (vii) Question no. **14** is a case-study based question with 2 sub-parts of **2** marks each.

SECTION A

Questions number 1 to 6 carry 2 marks each.

1. (a) Find : $\int \frac{1}{\sqrt{12 + 4x - x^2}} dx$ 2

OR

(b) Find : $\int \frac{xe^x}{(x + 4)^5} dx$ 2

2. Find the general solution of the following differential equation : 2

$$(4 + y^2)(3 + \log x) dx + x dy = 0$$

3. If $|\vec{a}| = 3$, $|\vec{b}| = 2\sqrt{3}$ and $\vec{a} \cdot \vec{b} = 6$, then find the value of $|\vec{a} \times \vec{b}|$. 2

4. Find the direction cosines of a line whose cartesian equation is given as $3x + 1 = 6y - 2 = 1 - z$. 2

5. In a toss of three different coins, find the probability of coming up of three heads, if it is known that at least one head comes up. 2



6. Two rotten apples are mixed with 8 fresh apples. Find the probability distribution of number of rotten apples, if two apples are drawn at random, one-by-one without replacement. 2

SECTION B

Questions number 7 to 10 carry 3 marks each.

7. Evaluate : $\int_0^{\pi/3} |\cos 3x| dx$ 3

8. (a) Find the general solution of the following differential equation : 3

$$2x e^{y/x} dy + (x - 2y e^{y/x}) dx = 0$$

OR

- (b) Find the particular solution of the differential equation $(2x^2 + y) \cdot \frac{dx}{dy} = x$; given that $y = 2$ when $x = 1$. 3

9. Using vectors, find the area of the triangle with vertices $A(-1, 0, -2)$, $B(0, 2, 1)$ and $C(-1, 4, 1)$. 3

10. (a) Find the shortest distance between the lines

$$\vec{r} = (\lambda + 1) \hat{i} + (\lambda + 4) \hat{j} - (\lambda - 3) \hat{k}, \text{ and}$$

$$\vec{r} = (3 - \mu) \hat{i} + (2\mu + 2) \hat{j} + (\mu + 6) \hat{k}. 3$$

OR

- (b) Find the distance of the point $P(4, 3, 2)$ from the plane determined by the points $A(-1, 6, -5)$, $B(-5, -2, 3)$ and $C(2, 4, -5)$. 3



SECTION C

Questions number 11 to 14 carry 4 marks each.

11. Find : $\int \frac{x^2 + x + 1}{(x + 1)(x^2 + 4)} dx$ 4

12. (a) Find the area bounded by the ellipse $x^2 + 4y^2 = 16$ and the ordinates $x = 0$ and $x = 2$, using integration. 4

OR

- (b) Find the area of the region $\{(x, y) : x^2 \leq y \leq x\}$, using integration. 4

13. Find the coordinates of the point where the line through $(4, -3, -4)$ and $(3, -2, 2)$ crosses the plane $2x + y + z = 6$. 4

Case-Study Based Question

14. A laboratory blood test is 98% effective in detecting a certain disease when it is in fact, present. However, the test also yields a false positive result for 0.4% of the healthy person tested. From a large population, it is given that 0.2% of the population actually has the disease.

Based on the above, answer the following questions :

- (a) One person, from the population, is taken at random and given the test. Find the probability of his getting a positive test result. 2
- (b) What is the probability that the person actually has the disease, given that his test result is positive ? 2